12.4 Permutations and Combinations

A **permutation** counts ordered arrangements — position matters, so swapping two chosen items creates a different outcome: ${}_nP_r = \dfrac{n!}{(n-r)!}$. A **combination** counts unordered selections — swapping two chosen items gives the *same* outcome (a committee is a committee no matter what order you name its members): ${}_nC_r = \dfrac{n!}{r!(n-r)!} = \dfrac{{}_nP_r}{r!}$. The single most important skill here is deciding *which one applies* before computing anything — ask "does order/role matter?" This worksheet also combines permutations/combinations with the Fundamental Counting Principle, restrictions (a specific person must/must not be included or placed), and probability.

Examples

Example 1: Factorial

Evaluate $6!$.

Solution

Identify: $n!$ means multiply all whole numbers from n down to 1.

Compute: $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$.

Answer: 720.

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Example 2: Permutation — arranging a subset in order

In how many ways can 3 of 7 different books be arranged on a shelf (order matters)?

Solution



Three ordered slots: 7 choices for the first, then 6, then 5 — order matters.

Identify: order matters (a different arrangement of the same 3 books is a different outcome), so this is a **permutation**. Use ${}_nP_r = \dfrac{n!}{(n-r)!}$ with $n = 7, r = 3$.

Compute: ${}_7P_3 = \dfrac{7!}{4!} = \dfrac{7 \times 6 \times 5 \times 4!}{4!} = 7 \times 6 \times 5 = 210.$

Answer: 210 ways.

Example 3: Combination — the same books, but order doesn't matter

In how many ways can 3 of the same 7 books be *chosen* (not arranged — order doesn't matter)?

Solution



The 3 circled books form one set — no matter what order they're named, it's the same selection.

Identify: order no longer matters (choosing books X, Y, Z is the same outcome regardless of the order named), so this is a **combination**. Use ${}_nC_r = \dfrac{n!}{r!(n-r)!}$ with $n = 7, r = 3$.

Compute: ${}_7C_3 = \dfrac{7!}{3!\,4!} = \dfrac{7 \times 6 \times 5}{3 \times 2 \times 1} = \dfrac{210}{6} = 35.$

Notice this equals $\dfrac{{}_7P_3}{3!} = \dfrac{210}{6} = 35$ — combinations always undercount permutations by the $r!$ orderings that get treated as "the same" selection.

Answer: 35 ways.

Example 4: P or C? — Officers with distinct roles

A club of 10 people elects a president, vice-president, and secretary. How many ways can these three distinct roles be filled?

Solution

The roles are different, so swapping who gets which role creates a genuinely different outcome — this is a **permutation**.

$$_{10}P_3 = \frac{10!}{7!} = 10 \times 9 \times 8 = 720.$$

Answer: 720 ways.

Example 5: P or C? — An unranked committee

The same club of 10 people chooses 4 people (with no distinct roles) for a committee. How many ways?

Solution

No roles to distinguish the four — this is a **combination**.

$$_{10}C_4 = \frac{10!}{4!\,6!} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = \frac{5040}{24} = 210.$$

Answer: 210 ways.

Example 6: Why $0!=1$

Explain why $0!$ is defined to equal 1, not 0.

Solution

The factorial pattern is $n! = n \times (n-1)!$. Working this backward from $n = 1$: $1! = 1 \times 0!$. Since $1! = 1$, we need $0! = 1$ to keep the pattern consistent.

It also matches counting logic: there is exactly 1 way to arrange zero objects (do nothing) — an empty arrangement is still one valid outcome, not zero outcomes.

Answer: $0! = 1$, by the recursive pattern and by counting logic (one way to arrange nothing).

Example 7: Reversing the relationship

If ${}_nC_3 = 20$, find ${}_nP_3$.

Solution

Use ${}_nP_r = {}_nC_r \times r!$:

${}_nP_3 = 20 \times 3! = 20 \times 6 = 120.$

Answer: 120.

Example 8: Permutation with a fixed position (restriction)

5 runners finish a race; medals go to the top 3 finishers in order (1st, 2nd, 3rd). If a particular runner, Maria, is guaranteed to finish 1st, in how many ways can the remaining medal positions (2nd and 3rd) be filled?

Solution

Maria's spot is fixed, so only the other 4 runners compete for the 2 remaining ordered positions:

${}_4P_2 = 4 \times 3 = 12.$

Answer: 12 ways.

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Example 9: Fundamental Counting Principle + permutations — license plates

A license plate has 3 different letters (no repeats) followed by 3 different digits (no repeats). How many plates are possible?

Solution



Letter slots (top) shrink 26→25→24; digit slots (bottom) shrink 10→9→8. Multiply the two stages together.

Letters: choose and arrange 3 of 26 letters, order matters (position matters on a plate): ${}_{26}P_3 = 26 \times 25 \times 24 = 15{,}600$.

Digits: choose and arrange 3 of 10 digits: ${}_{10}P_3 = 10 \times 9 \times 8 = 720$.

Combine with the Fundamental Counting Principle (multiply independent stages): $15{,}600 \times 720 = 11{,}232{,}000$.

Answer: 11,232,000 possible plates.

Example 10: Combination with a required member

A committee of 5 is chosen from 12 people, and Alex must be on it. How many committees are possible?

Solution

Since Alex's spot is guaranteed, only 4 more people need to be chosen — from the remaining 11 people, order doesn't matter:

$${}_{11}C_4 = \frac{11!}{4!\,7!} = \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1} = \frac{7920}{24} = 330.$$

Answer: 330 committees.

Example 11: Capstone — combinations and probability together

A lottery requires picking the correct 6 numbers from 49 (order doesn't matter). Given that ${}_{49}C_6 = 13{,}983{,}816$, find the probability of winning with one ticket.

Solution

There is exactly 1 winning combination out of all ${}_{49}C_6$ equally likely combinations:

$$P(\text{win}) = \frac{1}{{}_{49}C_6} = \frac{1}{13{,}983{,}816}.$$

Answer: $\dfrac{1}{13{,}983{,}816}$ — an illustration of how quickly combinations grow, and why lottery odds are so small.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Factorial

Evaluate $4!$.

Answer: **Verification:**

Problem 2: Permutation

Find ${}_6P_3$.

Answer: **Verification:**

Problem 3: Combination

Find ${}_8C_3$.

Answer: **Verification:**

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Problem 4: P or C?

The finishing order of 5 of 12 runners in a race: permutation or combination?

Answer: **Verification:**

Problem 5: P or C?

Choosing 5 of 12 runners for a relay team (no order): permutation or combination?

Answer: **Verification:**

Problem 6: Evaluate

Find ${}_4P_4$.

Answer: **Verification:**

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Problem 7: Evaluate

Find ${}_5C_1$.

Answer: **Verification:**

Problem 8: Evaluate

Find ${}_9C_2$.

Answer: **Verification:**

Problem 9: Evaluate

Find ${}_8P_2$.

Answer: **Verification:**

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Problem 10: Zero factorial

Evaluate $0!$.

Answer: **Verification:**

Problem 11: Relationship

If ${}_nC_3 = 20$, find ${}_nP_3$.

Answer: **Verification:**

Problem 12: Arrangement

In how many ways can 4 of 6 different novels be arranged on a shelf?

Answer: **Verification:**

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Problem 13: Selection

In how many ways can 4 of 6 people be chosen for an unranked group?

Answer: **Verification:**

Problem 14: Restriction

6 people sit in a row. Two specific people, X and Y, must sit at the two ends (in either order). How many seatings are possible?

Answer: **Verification:**

Problem 15: Challenge

A lottery requires picking 5 correct numbers from 40 (order doesn't matter), where ${}_{40}C_5 = 658{,}008$. Find the probability of winning with one ticket.

Answer: **Verification:**

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✓ Answer Key

1. 24
2. 120
3. 56
4. Permutation
5. Combination
6. 24
7. 5
8. 36
9. 56
10. 1
11. 120
12. ${}_6P_4 = 360$
13. ${}_6C_4 = 15$
14. $2 \times 4! = 48$
15. $\frac{1}{658{,}008}$

Great work — keep practising!